

Emergency Facility Location of Ambulances Using K-Means Clustering and Minimax



Ramkrishna Bharsakade , Sejal More, Sharwari Nandeshwar, Rahul Narnaware, and Raj Patil

Abstract The rational location of emergency medical services could play a crucial part in saving patient's life by improving access to the facilities. Every year, an increasing number of individuals lose their lives in accidents. One of the main reasons for the increase in mortality is the lack of access to medical facilities on nearby premises. This study aims to locate an optimal location for ambulances near the city's accident-prone zones, thereby reducing the waiting time of ambulances. The faster the ambulance reaches the accident spot, the quicker a patient will be able to reach the hospital. This study used K-means clustering and the minimax method to locate ambulances at optimal places. Using K-means clustering, clusters for accident-prone zones are formed; according to the value of k accident spots nearby, each other is grouped or clustered together. Finally, using minimax, coordinates for optimal location are calculated. Thus, after a complete study, two optimal locations were found that are most suitable for the location of the ambulance. Thus, the optimal location of an ambulance can help reduce the number of deaths due to road accidents.

Keywords Emergency medical services · K-means clustering · Minimax method

1 Introduction

Sources say that in India, accidents occur every two minutes. In 2020, 116,498 road accidents occurred on the National Highways, including the expressway, causing 47,984 deaths. As described, over 137,191 accidents were reported in 2019, in which 53,872 lost their lives, and many people suffered injuries. In Maharashtra, 24,941 accidents occurred in 2020, of which 10,773 were fatal accidents, and 11,569 got killed. Out of this, it is estimated that most of the accidents happen on National Highway or expressway. According to world traffic statistics, India has the most significant number of road accidents among 199 countries, followed by China and the United States in the second and third places. According to the WHO Global

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A. K. Shukla et al. (eds.), *Recent Advances in Mechanical Engineering*, Lecture Notes in Mechanical Engineering, https://doi.org/10.1007/978-981-99-1894-2_54

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Report, India is responsible for 11% of all global accidents. As a result, highways account for more than half of all accidents, and this study aims to cover one such highway and implement a system for that region so that deaths due to accidents can be prevented in that region. For this study, the region considered is Nagpur-Amravati Highway (NH-53). It covers an entire stretch of 75 km from Nagpur Chokardhani to Sarwadi-Khursapar Stretch. Most of the accidents occurring in this stretch are mainly due to overspeeding, overtake negligence (wrong lane and sides), no reflectors on trucks, potholes, tire-burst, trees, etc. And these accidents occur at specific locations at a high frequency. Also, increased fatalities in accidents are caused because the ambulance does not reach the accident spot on time. Thus, our goal is to locate ambulances at such optimal locations, reducing the overall time required for an ambulance to arrive and further reach the hospital. Our effort will be to bring fruitful improvements to the existing system to improve overall efficiency.

The paper is organized as follows: Sect. 2 discusses different research studies done in the field of emergency facility location under literature review, Sect. 3 will brief about the data collection required for the project. Section 4 discusses the implementation of K-means clustering and the minimax method. Section 5 contains results after implementation of the methodology. Section 6 discusses conclusion and future scope.

2 Literature Review

A generous amount of research has been done in the domain of facility location. Various models have been designed and presented over the past few years. A system proposed in [1] has discussed integrating vehicular ad-hoc network (VANET) along with navigation with Indian constellation (NAVIC). Here, a chip supporting VANET technology is fitted in every vehicle. This will send distress signals as soon as accidents occur, and to know the exact location of the accident, NAVIC is used. Connecting advanced and efficient IoT devices can make it possible to know the accident location in the least possible time. In [2], they designed a new protocol for preventing and mitigating risks. They propose a three-phase traffic accidents reduction strategy (TARS) system that predicts accidents in advance and directs traffic to avoid traffic jams that could cause accidents. Key features of the system are smart vehicles, road side units, trusted authorities (TA), and government officials (GA). The three phases in the proposed plan are as follows: a. The set stage where RSU periodically broadcasts its identity and location around it so that vehicles traveling at its range can communicate with it. Collects the location, speed, and direction of vehicles in their range. b. Authorization phase and c. performance phase, TARS focuses on using a distance measurement method to generate a warning message to prevent accidents and redirect traffic more efficiently.

Artificial intelligence has brought significant technological advancement in all possible fields/domains. Thus, implementing AI would help increase the efficiency of the system. For instance, if an ambulance carries patient suffering from cardiac

arrest now, the ambulance will take the patient to the nearest hospital. Still, the chances of survival are 20% if the hospital is not a hospital dealing with cardiac arrest. Thus, an AI can suggest the nearest cardiac system to the ambulance, i.e., to people working in EMS, thereby increasing the patient's chances of survival. In [3], Zvi Drezner, Avram Mehrez, and George O. Wesolowsky consider the minisum and minimax single facility location problems. In a specific case, if active service distance reaches or surpasses a particular value, the problem becomes a minisum or minimax single facility site problem. To solve minimax problems, certain computer graphics like Brady and Rosenthal are used. Emergency facility location problems can be solved using Tabu Heuristic. The Tabu search algorithm is employed in the Istanbul earthquake readiness scenario in [4], with extensive comparative analysis of different results related to a connection failure, independent link failure, and cases of link failure. It suggested that a model-dependent link failure improves the percentage of the most covered coverage.

While many research papers present different models on facility location, few of them have concentrated their attention on facility location for long-scale medical emergency situations (LEMS). In a research paper presented by Jia et al. [5], their primary goal was to study and discuss various conventional facility location models, including p-center and p-median. While designing a model for LEMS, characteristics of large-scale medical emergencies have been studied. According to different objectives, different models for LEMS have been formulated.

The Capacitated Facility Location Problem (CFLP) is developed in [6] as a cluster-based location assignment method. This method approximates the ideal placement and coverage of a set of facilities to meet the demand at multiple locations. The proposed approach consists of two steps: Allocating demand to facilities considering capacity constraints and minimizing costs, and iteratively optimizing the locations of facilities using a customized K-Means clustering method. In [7] provides brief insights in minimax problems. It describes about minimax problems with constraints like total average cost and distance travelled and propose a optimized solution. In [8], Francis RL studied the problem of finding a new location for a facility in the network that minimizes the maximum distance between the new facility and the existing facilities, using a network model of a system of transport links with nodes representing locations of existing facilities. A number of properties of the problem are developed that lead to a new solution for solving the problem when the network is a tree, and to a completely new equivalent spanning tree problem for networks in general. In [9], to minimize the distance between a mentioned number of (emergency) facilities and their assigned fixed-demand locations, the m-center problem is treated, where these facilities are placed anywhere along the road network. The problem is simulated using integer programming and successfully solved using a binary search technique and a combination of exact tests and heuristics. In [10], a problem of locating a single ambulance facility in a polygonal area X is discussed. The objective is to locate an ambulance station in such a way that the maximum distance of the route leading from the station to the hospital via the accident site is as small as possible. They mainly focus on A-distance, which Widmayer et al. introduced as a generalization of straight-line distance, and show how the errand boy problem

with A-distance can be used to solve PM. In [11], the accident-prone regions along expressways are analyzed to determine optimal locations for dispatching ambulance services (EMS). For this purpose, the mean-shift algorithm was used to determine the clusters and their centers and to locate a second base within the primary clusters. To ensure that dispatched ambulances arrive at the scene of an accident within 8 minutes, the locations of the auxiliary bases were created to complement the primary stations of EMS and to distribute the load. In [12], the authors discuss the application of AI in healthcare and various algorithms that are useful in this field. Various AI devices such as ML and NLP and related algorithms are mentioned, including the clustering algorithm. In [13], various criteria for the use of emergency departments are discussed. A fuzzy set approach is used to eliminate variance and uncertainty in decision making. In [14] discusses set covering problem in emergency service location and further comments on its practical applications and limitations. Further author formulates Hierarchical objective set covering which covers a large extent and considers responses within districts. In [15], another method was proposed to solve the location of emergency facilities. Here, the program was formulated as a set problem and solved as a linear programming problem.

3 Data Collection

As mentioned, the region of Nagpur-Amravati Highway (NH-53) has been considered for the study. It covers a total stretch of 75 km from Nagpur Chokardhani to Sarwadi-Khursapar Stretch. In this stretch of 75 Kms, two toll booths are located. The first one is Gondkhairi/ Kondhali Toll, and the second one is Karanja Ghadge Toll. One ambulance and one crane each are located at both the tolls. A survey was undertaken to understand the traffic details of the considered area (refer to Table 1.). Presented below are some of the observations collected while undertaking the survey.

Table 1 Traffic data collected during survey

Sr. No	Parameters	Observations
1	Daytime traffic: heading toward Nagpur	In-6586 vehicles; out-7520 vehicles (Total vehicles: 14,103)
2	Average vehicle type occupancy	Two-wheelers (1.75%), car (3.15%), auto (4.19%), and taxi (4.02%)
3	Frequency of trips (private vehicles)	Daily-45% and weekly-14%
4	Purpose of the trip (private vehicles)	Work-32%, business-27%, and education-10%
5	Frequency of trips (goods vehicles)	Daily-46% and weekly-30%
6	Purpose of trips (good vehicles)	Loading-45% and unloading-52%

Most of the accidents occurring in this stretch are mainly due to overspeeding, overtake negligence (wrong lane and sides), no reflectors on trucks, potholes, tire-burst, trees, etc. And these accidents occur at specific locations at a high frequency; those locations are black spots. Thus, such black spots or accident-prone zones were identified for this region. There are 3 primary health centers (PHCs) in the region, located at Kondhali, Bajargaon, and Karanja. And there are 3 nearest private hospitals located. They are (1) Nakshatra Multispecialty Hospital, (2) Gajbe Hospital, and (3) Welltreat Multispecialty Hospital. Along with these hospitals, Mayo Hospital, which is a Government Hospital located in central Nagpur, is considered. Data regarding accident spots (refer to Table 2) and nearest hospital (refer to Table 3) have been collected through surveys.

Further, using techniques like K-means and minimax on the coordinates of accident-prone zones, two optimal locations were identified.

Table 2 Coordinates for accident spots

Accident spots	X	Y
8th Mile	21.144212	78.975467
Waddhamna	21.140750	78.950247
Chokar Dhani Divider	21.136884	78.919868
Black spot 265 Bajargaon bus stop	21.137012	78.773755
Black spot 264 Junapani bus stop (Kondhali)	21.151843	78.567808
Oriental plaza Karanja	21.155492	78.382767
Karanja ghadge bus stop	21.160994	78.408981
Satnav	21.140831	78.808513

Table 3 Coordinates for nearest hospitals

Nearest hospital	x	y
Asha Hospital Dattwadi	21.151767	78.972245
Nakshatra multispecialty hospital, wadi	21.322239	78.972472
PHC Gondkhairi	21.137696	78.907264
PHC Kondhali	21.142665	78.632036
Madhavbaug	21.158016	78.613154
Rural Hospital	21.167853	78.410795

4 Methodology

4.1 K-Means Clustering

K-means clustering is a centroid-based algorithm used to cluster unlabeled data. The value of k , i.e., the number of clusters, is specified. Here, the value of k , the number of clusters to be made, is already defined, or an optimal number of clusters is evaluated by using various techniques like the elbow method, silhouette method, gap statistic, etc. Using the provided data, the value of k -specific points is chosen as centroids, and the distance of all points in the dataset is calculated. The points closest to centroids will form a cluster (Fig. 1).

Using an algorithm for K-means clustering, 8 coordinates were clustered for accident spots into two groups. According to the result, the first cluster consists of a set of 5 locations, while the second cluster consists of three locations. While setting a suitable value for k , three input values for the k were provided, i.e., $k = 2$, $k = 3$, and $k = 4$. This algorithm is executed using RStudio wherein various packages like dplyr, VIM, facto extra, gridExtra, and cluster are used. The following is the flow of the code written for the execution of this method (as shown in Fig. 2).

For evaluation of optimal clusters, gap statistic method is applied, in which the value of k will be optimal and appropriate. One of the traditional approaches for calculating the number of clusters in a dataset is to use the gap statistic. It compares the log (Wk) graph to its expectation, given the data's acceptable null reference distribution to quantify and standardize it. Wk is the within-cluster dispersion.

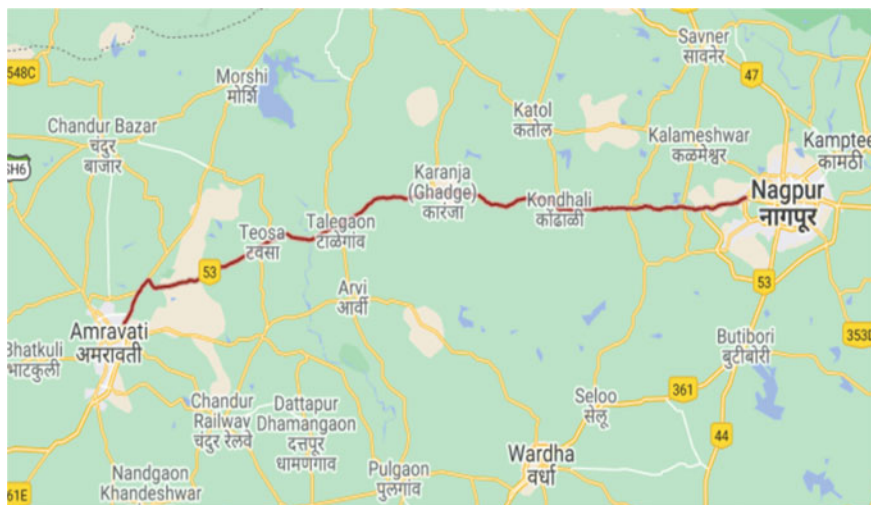
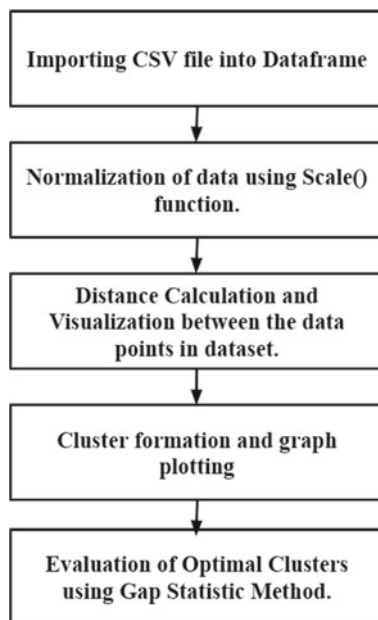


Fig. 1 Nagpur-Amravati highway (Source: Google Maps)

Fig. 2 Steps for performing K-means clustering



4.2 Minimax Method

The minimax approach refers to minimizing the maximum distance between the new facility and any existing facilities. Rectilinear distance is considered between devices while using minimax. A rectilinear distance is a distance associated with orthogonal or perpendicular lines to each other, often used to measure the distance to a city grid. Suppose coordinates for two facilities are given as (x_1, y_1) and (x_2, y_2) , and the rectilinear distance is given [16]:

$$D = |x_1 - x_2| + |y_1 - y_2| \quad (1)$$

Minimax Model

The minimax formula is given by:

$$\begin{aligned}
 c_1 &= \min \{a_i + b_i - h_i\} \\
 c_2 &= \max \{a_i + b_i + h_i\} \\
 c_3 &= \min \{-a_i + b_i - h_i\} \\
 c_4 &= \max \{-a_i + b_i + h_i\} \\
 c_5 &= \max \{c_2 - c_1, c_4 - c_3\}
 \end{aligned}$$

Set of optimal solutions: line segment defined by the following end points:

$$(x_1^*, y_1^*) = 1/2(c_1 - c_3, c_1 + c_3 + c_5)$$

$$(x_2^*, y_2^*) = 1/2(c_2 - c_4, c_2 + c_4 - c_5)$$

In our case, we have used minimax facility location for calculating the optimal location of an ambulance concerning six (known) possible accident locations whose coordinates are

$$A1 = (21.144212, 78.975467)$$

$$A2 = (21.140750, 78.950247)$$

$$A3 = (21.136884, 78.567808)$$

$$A4 = (21.137012, 78.773755)$$

$$A5 = (21.151843, 78.567808)$$

$$A6 = (21.155492, 78.382767)$$

$$A7 = (21.160994, 78.408981)$$

$$A8 = (21.140831, 78.808513)$$

where A1, A2, A3, A4, A5, A6, A7, and A8 are accident spots as shown in Table 2. Similarly, the possible (known) coordinates of the nearest hospital from the accident spots are

$$B1 = (21.151767, 78.972245)$$

$$B2 = (21.322239, 78.972472)$$

$$B3 = (21.137696, 78.907264)$$

$$B4 = (21.142665, 78.632036)$$

$$B5 = (21.158016, 78.613154)$$

$$B6 = (21.158016, 78.410795)$$

B1, B2, B3, B4, B5, and B6 are the nearest hospitals to the accident spots, as shown in Table 2. The goal is to reduce the maximum distance from the ambulance location to the accident spot and from the accident spot to the nearest hospital. To do this, the distances from the mishap sites to their nearby hospitals are required, i.e., the rectilinear distance, let us say it “H” by applying the formula (1) [17]. The calculations are done in MS Excel using standard procedures for the minimax method.

5 Results

After implementing K-means clustering (as shown in Fig. 3), two sets of clusters were obtained. The first one consists of 5 locations: 8th Mile, Waddhamana, Choker

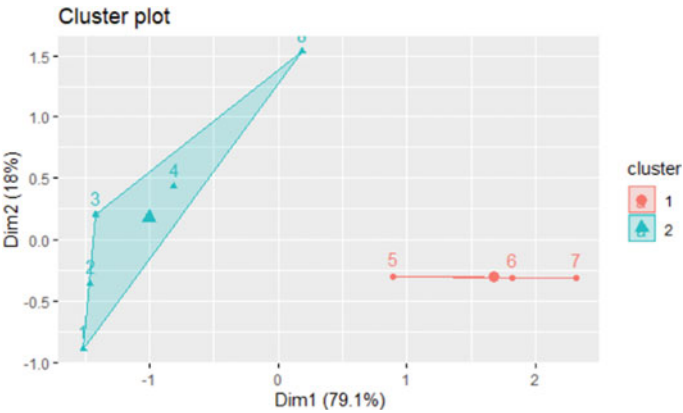


Fig. 3 Clustering of accident spots using K-means clustering



Fig. 4 Minimax location for Cluster 1 and Cluster 2

Dhani Divider, Bajargaon bus stop, and Satnavri. The second cluster consists of 3 locations: Junapani bus stop, Oriental Plaza Karanja, and Karanja Ghadge bus stop.

The results obtained after applying the minimax formula give two optimal locations for the ambulance for the stretch of 8th Mile to Satnavri (19 km) which is on the line connecting these two points, i.e., $(X1^*, Y1^*) = (21.14075, 78.7555)$ and $(X2^*, Y2^*) = (21.14075, 78.7408)$, and for the stretch from Bazargaon bus stop to Karanja Ghadge bus stop(40 km), optimal location for an ambulance is on the line connecting these two points $(X1^*, X2^*) = (21.15549, 78.64099)$ and $(X2^*, Y2^*) = (21.13701, 78.6225)$ (as shown in Fig. 4). After implementing the given results, time calculations can be done to evaluate how much time it saves after placing ambulances in the obtained optimal locations.

6 Conclusion and Future Scope

Thus, using K-means clustering and minimax optimal location for ambulances are found. Locating ambulances in these places might save time to reach the hospital, which can ultimately help to bring down the mortality rate. In future, implementing the findings in this project in real life, i.e., placement of ambulances on obtained

optimal locations on the highway can be calculated and the approximate time required to reach the hospital from the location. Further comparing results from this model and the previous model, the efficiency of the result can be evaluated.

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